

Evaluation of Racial, Ethnic, and Socioeconomic Associations With Treatment and Survival in Uveal Melanoma, 2004-2014

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IMPORTANCE Identifying disparities in uveal melanoma (UM) treatment patterns and survival across racial, ethnic, and socioeconomic (SES) groups reveals possible inequities in ophthalmologic health care.

OBJECTIVE To examine the association of race, ethnicity, and SES with UM treatment and survival.

DESIGN, SETTING, AND PARTICIPANTS A retrospective cohort analysis of 28% of the US population using the Surveillance, Epidemiology, and End Results (SEER) 18 registries from January 1, 2004, to December 31, 2014, was conducted. Data analysis was performed from April to July 2018. SEER identified 4475 individuals using *International Classification of Diseases for Oncology, Third Edition* site and morphology codes.

EXPOSURES Race, ethnicity, and SES estimated by tertile using Yost Index composite scores.

MAIN OUTCOMES AND MEASURES Treatment odds ratios (ORs), 1-year and 5-year survival estimates, mortality hazard ratios (HRs), and Kaplan-Meier survival curves. Hypothesis was formulated before data collection.

RESULTS Multivariate analyses of 4475 individuals (2315 [51.7%] men; non-Hispanic white, 4130 [92.3%]; nonwhite, 345 [7.7%]) showed that patients who were nonwhite (OR, 1.45; 95% CI, 1.12-1.88) and socioeconomically disadvantaged (lower SES: OR, 2.21; 95% CI, 1.82-2.68; middle SES: OR, 1.86; 95% CI, 1.56-2.21) were more likely to receive primary enucleation. No interactions were observed between race/ethnicity, SES, and stage at diagnosis. From 2004 to 2014, rates of primary enucleation decreased across all racial/ethnic and SES groups, but disparities persisted. Socioeconomically disadvantaged patients had lower 5-year all-cause survival rates (lower SES: 69.2%; middle SES: 68.1%; and upper SES: 73.8%), although disease-specific survival did not vary significantly by racial/ethnic or SES strata. Mortality risk was associated with older age at diagnosis (56-68 years: HR, 1.70; 95% CI, 1.44-2.01; ≥ 69 years: HR, 3.32; 95% CI, 2.85-3.86), advanced stage of UM (stage 2: HR, 1.40; 95% CI, 1.19-1.65; stage 3: HR, 2.26; 95% CI, 1.87-2.73; and stage 4: HR, 10.09; 95% CI, 7.39-13.77), and treatment with primary enucleation (HR, 2.14; 95% CI, 1.88-2.44) with no racial/ethnic or SES variation.

CONCLUSIONS AND RELEVANCE In this study, SEER data from 2004 to 2014 suggest that nonwhite and socioeconomically disadvantaged patients with UM are more likely to be treated with primary enucleation, although no such variation appears to exist in disease-specific survival. These differences reveal opportunities to address issues regarding treatment choice in UM.

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As shown in various oncologic conditions, patients from disadvantaged backgrounds often face barriers to care from poor follow-up to delayed diagnosis to difficulty accessing appropriate treatment.¹⁻⁴ Overall, research has shown that there are many clinically important differences between patients with uveal melanoma (UM) of different races, ethnicities, and socioeconomic status (SES).⁵ The age-adjusted incidence of UM differs by race (0.31 in black, 0.38 in Asian, 1.67 in Hispanic, and 6.02 in non-Hispanic white populations),⁶⁻¹⁰ while the prevalence of choroidal nevi, which have a known association with UM development,¹¹ is 0.6% in black, 2.7% in Hispanic, and 5.6% in non-Hispanic white populations.¹² One study suggested decreased mortality in Hispanic compared with white patients (odds ratio [OR], 0.6),¹³ while another study found no statistically significant difference in metastasis, survival, or mortality between ethnicities.¹ Histopathologic studies have reported that the clinical and genetic course of non-white patients with UM may be different than in white patients, with Asian Indian individuals presenting younger,¹⁴ Mexican-Mestizo patients showing a previously unreported polymorphism,¹⁵ and Hispanic patients presenting with unique symptoms and broader-based tumors.² Various case reports have shown that UM occurs in all races.^{2,16-20}

Research into the association of SES with UM incidence and treatment has been inconclusive. Zinkhan et al²¹ reported a decreased incidence in higher SES populations, suggesting possible occupational risk, while Shildkrot et al²² found the opposite, even when controlling for race; 2 other studies found no association between SES and UM incidence.^{23,24} Lockington et al²⁵ looked for differences in treatment patterns by SES, finding no statistically significant difference. These varying results suggested a need for greater power and validated SES measures in future studies.

It is unclear if there are patient-level factors, other than clinical presentation, that would influence treatment selection. In retinoblastoma, the Surveillance, Epidemiology, and End Results (SEER) 18 database has been used to examine racial, ethnic, and socioeconomic disparities in enucleation.^{26,27} However, to our knowledge, similar analyses have yet to be conducted in UM; this study addressed these gaps. We hypothesized that patients who are nonwhite and of lower SES experience increased adjusted odds of enucleation and worse survival.

Methods

Study Design and Population

This was a retrospective cohort study. The National Cancer Institute SEER 18 registries represented 28% of the US population between 2004 and 2014.^{28,29} SEER began with 7 registries in 1973, expanding over time to 18 registries in 2000. The Stanford University Institutional Review Board ruled that there was no need for their approval or for informed consent for this study because SEER data are available to the public with the limitations of their data use agreement.³⁰ All work was conducted in compliance with the tenets of the Declaration of Helsinki.³¹ Data analysis for this study was performed from

Key Points

Question Is race/ethnicity or lower socioeconomic status in patients with uveal melanoma associated with increased odds of primary enucleation or decreased survival?

Findings This cohort study of 4475 patients with uveal melanoma from 2004 to 2014 found that patients who are nonwhite and socioeconomically disadvantaged are more likely to receive primary enucleation despite disease stage at presentation. There appear to be no racial, ethnic, or socioeconomic differences in disease-specific survival.

Meaning The findings of this study suggest that racial, ethnic, and socioeconomic status disparities are present in uveal melanoma treatment and survival; further research is needed to elucidate the sources of these disparities.

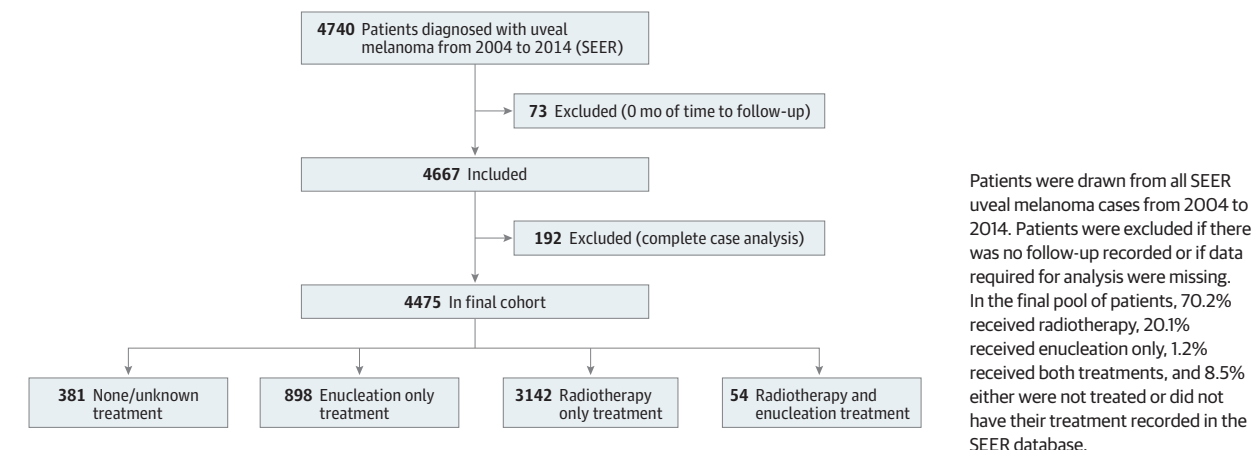
April to July 2018. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline for cohort studies. Hypothesis was formulated before data collection.

Individuals with UM that was diagnosed between January 1, 2004, and December 31, 2014, were identified using *International Classification of Diseases for Oncology, Third Edition* site (C69.3: choroid, C69.4: iris and ciliary body, and C69.2: retina) and morphology (8720/3-8780/3) codes.³² As discussed by Aronow et al,⁸ primary melanoma typically does not originate from the retina; therefore, the term *retinal melanoma* likely represents miscoding of UM. Cases with site code C69.2 were accordingly included in this analysis. The 2004-2014 time frame was chosen to allow use of the American Joint Committee on Cancer (AJCC) 7th edition stage data that SEER began regularly recording for ocular cancers in 2004. Individuals with zero months of time to follow-up were excluded (n = 73) (Figure).³³

Variables

This study examined 2 main exposures of interest: race/ethnicity and SES. The SEER database collected both race and ethnicity data for every patient in the study cohort. Race is reported as white, black, Asian, Pacific Islander, American Indian /Alaska Native, or other. Ethnicity is reported as Hispanic or non-Hispanic. Because UM is uncommon in non-white patients, all nonwhite and Hispanic patients were combined into one group for comparison against non-Hispanic white patients. In this study, SES was estimated using the Yost Index—a composite score that incorporates many variables related to SES (percentage of people below the federal poverty line, unemployment rate, median household income, percentage of households with <1 person per room, percentage of people with less than a high school education, and percentage of people with a bachelor's degree) in a patient's census tract.³⁴ SEER developed this index for all of the census tracts covered by its 18 registries.³⁵ The patients in each census tract were geocoded at the patient level and labeled as part of upper, middle, and lower SES groups. Previous studies have reported that census tract variables are more representative of SES than county-level

Figure. Inclusion of Patients With Uveal Melanoma From the Surveillance, Epidemiology, and End Results (SEER) Database



variables.³⁵ Census tracts are smaller and less economically diverse than counties, meaning individuals share more economic characteristics with people in their own tract than they do with those in their county.³⁶ The Yost Index also factors in education, occupation, and income.³⁷ In addition, the Yost Index has been used and validated in previous cancer studies.^{34,35,38} Other patient-level covariates included in this study were age at diagnosis, sex, and AJCC stage group at diagnosis. Age at diagnosis was divided into tertiles (lower, 0-55; middle, 56-68; and upper, ≥69 years).

This study examined 2 main outcomes: treatment status and survival. Treatment was coded as radiotherapy only, enucleation only, radiotherapy and enucleation, or no/unknown. In the SEER database, radiotherapy status is coded as either yes or no/unknown owing to incomplete reporting of radiotherapy to SEER registries. Given the fact that SEER radiotherapy data have reasonable sensitivity (80%)³⁹ and radiotherapy is the standard care for UM, radiotherapy data were included.⁴⁰ Radiotherapy status was coded as yes if any form of radiotherapy was received. Enucleation status was determined using SEER surgical codes as defined by Truong et al.²⁶ Patients who did not receive enucleation may have still undergone other less severe surgical procedures. Survival was evaluated at both the all-cause and disease-specific levels. In addition, treatment trends over time were studied.

All variables were missing less than 10% data except the AJCC stage group (26.3% missing). Missing stage group data were found to be missing at random, demonstrating a systemic association between the propensity of missing values and treatment status. However, given incomplete reporting of radiotherapy to SEER (radiotherapy status can be coded only as yes or no/unknown), imputation of stage data could not be performed. To avoid introducing misclassification bias, individuals missing stage data were not excluded. Therefore, a separate stage group category for those missing stage data was created and included in regression analysis. A complete case analysis was conducted excluding individuals missing data for any variable except stage group. Adults with primary tumors other than UM were not included in disease-specific survival

analysis, as SEER does not code for disease-specific vital status in that population (n = 756).

Statistical Analysis

To provide descriptive statistics, we calculated frequency counts and percentages and completed χ^2 analyses. Factors associated with treatment status were estimated using univariate and multivariate multinomial logistic regression with radiotherapy only as the reference group. The final model was composed of only covariates that showed greater than 10% confounding of SES or race/ethnicity. Kaplan-Meier curves stratified by race/ethnicity and SES were generated for all-cause and disease-specific survival and compared using the log-rank test. One-year and 5-year survival estimates were obtained. All-cause and disease-specific survival were also modeled using univariate and multivariate Cox proportional hazards regression. The final model similarly retained all covariates showing 10% or more confounding of race/ethnicity or SES. Changes in treatment trends over time were evaluated using the Cochran-Armitage χ^2 test for trend. Differences in treatment rates across racial, ethnic, and socioeconomic strata by time period were assessed using Pearson χ^2 test. All *P* values were calculated as 2-sided tests. A significance level of *P* < .05 was established for all tests. No adjustments were made for multiple testing. All analyses were completed using SAS, university version 9.4 (SAS Institute Inc).

Results

Population Characteristics

The study population included 4475 patients (Figure and Table 1) over the 11-year study period. Approximately half of the patients were male (2315 [51.7%]). The age range at diagnosis was 5 to 55 years (1539 [34.4%]), 56 to 68 years (1437 [32.1%]), and 69 years or older (1499 [33.5%]). Most tumors were either AJCC stage 1 (1137 [25.4%]) or stage 2 (1768 [39.5%]). The diagnosis of UM was confirmed by positive histologic test results (1858 [41.5%]), radiographic test findings (1030 [23.0%]), direct visualization (734 [16.4%]), or positive cytologic test re-

Table 1. Characteristics of Patients With Uveal Melanoma

				SES				
Characteristic	Non-Hispanic white	Nonwhite	P value	Lower	Middle	Upper	P value	Total, No. (%)
All patients	4130 (92.3)	345 (7.7)		978 (21.9)	1498 (33.5)	1999 (44.7)		4475 (100)
Age at diagnosis, tertile								
Lower (0-55 y)	1371 (33.2)	168 (48.7)	<.001	342 (35.0)	509 (34.0)	688 (34.4)	.28	1539 (34.4)
Middle (56-68 y)	1335 (32.3)	102 (29.6)		302 (30.9)	464 (31.0)	671 (33.6)		1437 (32.1)
Upper (≥69 y)	1424 (34.5)	75 (21.7)		334 (34.2)	525 (35.0)	640 (32.0)		1499 (33.5)
Sex								
Male	2155 (52.2)	160 (46.4)	.04	475 (48.6)	777 (51.9)	1063 (53.2)	.06	2315 (51.7)
Female	1975 (47.8)	185 (53.6)		503 (51.4)	721 (48.1)	936 (46.8)		2160 (48.3)
Race/ethnicity								
Non-Hispanic white	NA	NA		862 (88.1)	1394 (93.1)	1874 (93.7)	<.001	4130 (92.3)
Nonwhite	NA	NA		116 (11.9)	104 (6.9)	125 (6.3)		345 (7.7)
Socioeconomic status tertile								
Upper	NA	NA		NA	NA	NA		1999 (44.7)
Middle	NA	NA		NA	NA	NA		1498 (33.5)
Lower	NA	NA		NA	NA	NA		978 (21.9)
AJCC stage group								
Stage 1	1059 (25.6)	78 (22.6)	.60	227 (21.2)	379 (25.3)	531 (26.6)	.77	1137 (25.4)
Stage 2	1631 (39.5)	137 (22.6)		397 (40.6)	594 (39.7)	777 (38.9)		1768 (39.5)
Stage 3	545 (13.2)	44 (12.8)		132 (13.5)	205 (13.7)	252 (12.6)		589 (13.2)
Stage 4	61 (1.5)	6 (1.7)		15 (1.5)	21 (1.4)	31 (1.6)		67 (1.5)
Missing	834 (20.2)	80 (23.2)		207 (21.2)	299 (20.0)	408 (20.4)		914 (20.4)
Diagnostic confirmation								
Clinical diagnosis	268 (6.5)	23 (6.7)		51 (5.2)	102 (6.8)	138 (6.9)		291 (6.5)
Direct visualization	695 (16.8)	39 (11.3)		126 (12.9)	216 (14.4)	392 (19.6)		734 (16.4)
Positive								
Cytology	442 (10.7)	44 (12.8)	.055	80 (8.2)	144 (9.6)	262 (13.1)	<.001	486 (10.9)
Histology	1692 (41.0)	166 (48.1)		488 (49.9)	615 (41.1)	755 (41.0)		1858 (41.5)
Lab test/marker	12 (0.3)	0 (0.0)		0 (0.0)	6 (0.4)	6 (0.3)		12 (0.3)
Microscopic confirmation/unspecified	12 (0.3)	1 (0.3)		3 (0.3)	6 (0.4)	4 (0.2)		13 (0.3)
Radiography	961 (23.3)	69 (20.0)		223 (22.8)	388 (25.9)	419 (21.0)		1030 (23.0)
Unreported	48 (1.2)	3 (0.9)		7 (0.7)	21 (0.1)	23 (0.1)		51 (1.1)
Treatment status								
Enucleation only	804 (19.5)	94 (27.2)	.004	264 (27.0)	351 (23.4)	283 (14.2)	<.001	898 (20.1)
Radiation only	2923 (70.8)	219 (63.5)		625 (63.9)	1007 (67.2)	1510 (75.5)		3142 (70.2)
Both enucleation and radiation	48 (1.2)	6 (1.7)		10 (1.0)	15 (1.0)	29 (1.5)		54 (1.2)
Unknown/none	355 (8.6)	26 (7.5)		79 (8.1)	125 (8.3)	177 (8.9)		381 (8.5)
Vital status								
Dead from uveal melanoma	538 (13.0)	51 (14.8)	.001	132 (13.5)	215 (14.4)	242 (12.1)	.002	589 (13.2)
Alive or dead of other cause	2870 (69.5)	260 (75.4)		709 (72.5)	1048 (70.0)	1373 (68.7)		3130 (69.9)
Unreported	722 (17.5)	34 (9.9)		137 (14.0)	235 (15.7)	384 (19.2)		756 (6.9)

Abbreviations: AJCC, American Joint Committee on Cancer; NA, not applicable; SES, socioeconomic status.

sults (486 [10.9%]). Most patients received radiotherapy only (3142 [70.2%]) as treatment, while the remainder largely received enucleation only (898 [20.1%]); only 54 patients (1.2%) received both treatments. Data were missing on 381 patients (8.5%). At the time of analysis, 589 patients (13.2%) of the total cohort had died of UM.

Treatment

The study population was 92.3% (n = 4130) non-Hispanic white and 7.7% (n = 345) nonwhite (Table 1). While 94 nonwhite in-

dividuals (27.2%) underwent enucleation without radiotherapy, only 804 non-Hispanic white patients (19.5%) received primary enucleation. Most of the non-Hispanic white (2923 [70.8%]) and nonwhite (219 [63.5%]) cohorts were treated with radiotherapy only (Table 1) (P = .004).

Per the SEER-developed Yost Index, SES in the study cohort was categorized as lower (978 [21.9%]), middle (1498 [33.5%]), and upper (1999 [44.7%]) (Table 1). More patients (264 [27.0%]) in the lower tertile underwent enucleation without radiotherapy compared with those in the middle (351 [23.4%])

Table 2. Treatment Status in Patients With Uveal Melanoma Compared With Radiotherapy Only

Treatment	Characteristic		OR (95% CI)	
			Unadjusted	Adjusted
Enucleation	SES tertile	Upper	1 [Reference]	1 [Reference]
		Middle	1.86 (1.56-2.22) ^a	1.86 (1.56-2.21) ^a
		Lower	2.25 (1.86-2.73) ^a	2.21 (1.82-2.68) ^a
	Race/ethnicity	Non-Hispanic white	1 [Reference]	1 [Reference]
		Nonwhite	1.56 (1.21-2.01) ^a	1.45 (1.12-1.88) ^a
Both enucleation and radiation	SES tertile	Upper	1 [Reference]	1 [Reference]
		Middle	0.78 (0.41-1.45)	0.77 (0.41-1.45)
		Lower	0.83 (0.40-1.72)	0.81 (0.39-1.67)
	Race/ethnicity	Non-Hispanic white	1 [Reference]	1 [Reference]
		Nonwhite	1.67 (0.71-3.95)	1.70 (0.72-4.02)
Unknown/none	SES tertile	Upper	1 [Reference]	1 [Reference]
		Middle	1.06 (0.83-1.35)	1.06 (0.83-1.35)
		Lower	1.08 (0.81-1.43)	1.08 (0.82-1.43)
	Race/ethnicity	Non-Hispanic white	1 [Reference]	1 [Reference]
		Nonwhite	0.98 (0.64-1.49)	0.97 (0.64-1.48)

Abbreviations: OR, odds ratio;
SES, socioeconomic status.

^a Statistically significant difference.

Table 3. One-Year and 5-Year All-Cause and Disease-Specific Survival in Patients With Uveal Melanoma

Characteristic	Disease-specific survival, %			All-cause survival, %		
	1 y	5 y	P value	1 y	5 y	P value
All	97.6	81.4		95.9	70.9	
Race/ethnicity						
Non-Hispanic white	97.8	81.5	.51	95.9	70.8	.66
Nonwhite	96.1	80.2		95.8	72.0	
Socioeconomic status tertile						
Upper	98.3	83.7	.06	96.5	73.8	.003
Middle	96.6	79.7		94.9	68.1	
Lower	97.8	79.5		96.4	69.2	

and upper (283 [14.2%]) tertiles (Table 1). Most patients in the upper (1510 [75.5%]), middle (1007 [67.2%]), and lower (625 [63.9%]) tertiles received radiotherapy only ($P < .001$).

Multivariable multinomial regression revealed increased adjusted odds of receiving primary enucleation compared with receiving only radiotherapy for multiple sociodemographic and socioeconomic groups (Table 2). Patients who underwent enucleation without radiotherapy were more likely to be of nonwhite race/ethnicity (OR, 1.45; 95% CI, 1.12-1.88). Patients of middle SES (OR, 1.86; 95% CI, 1.56-2.21) or lower SES (OR, 2.21; 95% CI, 1.82-2.68) were also more likely to receive only enucleation compared with patients of upper SES. Race/ethnicity and SES were not associated with receiving both enucleation and radiotherapy compared with radiotherapy only. Sex, age at diagnosis, and AJCC stage did not confound the associations between treatment and race, ethnicity, and SES and were therefore not included in the final model. Interaction terms for effect modification between stage and race/ethnicity and SES were also not associated with treatment.

Across the study period from 2004 to 2014, overall rates for primary enucleation decreased from 26.9% to 16.5%, while rates for treatment with radiotherapy increased from 60.4% to 74.2% (eFigure 2A, eTable in the Supplement). Treatment

trends by race/ethnicity and SES were examined across 2 approximately equal time periods (2004-2009 and 2010-2014) (eFigure 2B and 2C in the Supplement). Rates for primary enucleation decreased over time across all racial/ethnic and socioeconomic categories (comparing 2004-2009 with 2010-2014; race/ethnicity: non-Hispanic white, 9.7% to 7.4%; $P < .001$; nonwhite, 9.3% to 6.0%, $P = .03$; SES: lower, 8.2% to 8.0%, $P = .045$; middle, 9.2% to 7.5%, $P = .23$; and upper, 10.8% to 6.7%; $P < .001$). Despite overall decreases in isolated enucleation rates, within each time period, nonwhite individuals consistently exhibited higher rates of primary enucleation than non-Hispanic white patients, as did patients of lower or middle SES compared with those of upper SES (2004-2009: race/ethnicity, $P = .01$; SES, $P < .001$; 2010-2014: race/ethnicity, $P = .11$; SES, $P < .001$).

Survival

During the study period, 589 patients (13.2%) died of UM (Table 1). Disease-specific 1- and 5-year survival were 97.6% and 81.4%, respectively (Table 3). Disease-specific survival estimates were similar across racial/ethnic groups (non-Hispanic white: 1 year, 97.8% and 5 years, 81.5%; nonwhite: 1 year, 96.1% and 5 years, 80.2%) and socioeconomic strata (lower: 1 year, 97.8% and 5 years, 79.5%; middle: 1 year, 96.6%

Table 4. Mortality in Patients With Uveal Melanoma

Characteristic	Mortality			
	Disease-specific		All-cause	
	HR (95% CI)	P value ^a	HR (95% CI)	P value ^a
Socioeconomic status tertile				
Upper	1 [Reference]		1 [Reference]	
Middle	1.07 (0.67-1.27)	.49	1.11 (0.98-1.27)	.12
Lower	0.96 (0.77-1.18)	.66	1.06 (0.91-1.23)	.91
Age at diagnosis, tertile				
Lower (0-55 y)	1 [Reference]		1 [Reference]	
Middle (56-68 y)	1.50 (1.22-1.84)	<.001	1.70 (1.44-2.01) ^b	<.001
Upper (≥69 y)	2.01 (1.64-2.45)	<.001	3.32 (2.85-3.86) ^b	<.001
Treatment status				
Radiotherapy only	1 [Reference]		1 [Reference]	
Enucleation only	3.07 (2.59-3.68)	<.001	2.14 (1.88-2.44) ^b	<.001
Radiotherapy and enucleation	3.86 (2.33-6.39)	<.001	1.77 (1.12-2.80) ^b	.01
Unknown/none	1.08 (0.76-1.49)	.65	1.20 (0.96-1.50)	.96
Race/ethnicity				
Non-Hispanic white	1 [Reference]		1 [Reference]	
Nonwhite	1.11 (0.83-1.48)	.48	0.99 (0.79-1.25)	.96
AJCC stage group ^c				
Stage 1	NA		1 [Reference]	
Stage 2	NA		1.40 (1.19-1.65) ^b	<.001
Stage 3	NA		2.26 (1.87-2.73) ^b	<.001
Stage 4	NA		10.09 (7.39-13.77) ^b	<.001
Missing	NA		1.48 (1.23-1.79) ^b	<.001

Abbreviations: AJCC, American Joint Committee on Cancer; HR, hazard ratio; NA, not applicable.

^a Differences calculated using Pearson χ^2 test.

^b Statistically significant difference.

^c Stage did not confound the associations between socioeconomic status, race, ethnicity, and survival in the disease-specific mortality analysis; however, it did so for the all-cause mortality analysis and so was included in the final all-cause mortality model.

and 5-year, 79.7%; and upper: 1 year, 98.3% and 5 years, 83.7%) ($P = .06$) (eFigure 1A in the [Supplement](#)).

All-cause 1- and 5-year survival were 95.9% and 70.9%, respectively (Table 3). All-cause survival estimates were similar across racial/ethnic strata (non-Hispanic white: 1 year, 95.9% and 5 years, 70.8%; nonwhite: 1 year, 95.8% and 5 years, 72.0%; $P = .66$) but varied across SES strata (lower: 1 year, 96.4%; and 5 years, 69.2%; middle: 1 year, 94.9% and 5 years, 68.1%; and upper: 1 year, 96.5% and 5 years, 73.8%; $P = .003$) ($P = .003$) (eFigure 1B in the [Supplement](#)).

Multivariable Cox proportional hazards regression revealed increased adjusted disease-specific mortality for multiple sociodemographic and clinical groups (Table 4). Patients diagnosed at an older age experienced increased adjusted disease-specific mortality (middle tertile 56-68 years: hazard ratio [HR], 1.50; 95% CI, 1.22-1.84; upper tertile ≥69 years: HR, 2.01; 95% CI, 1.64-2.45) compared to the youngest tertile. Hazard ratios were also increased for patients treated with enucleation only (HR, 3.07; 95% CI, 2.59-3.68) and for those treated with both radiotherapy and enucleation (HR, 3.86; 95% CI, 2.33-6.39) compared to those treated with radiotherapy only. Race/ethnicity and SES were not associated with mortality. Sex and AJCC stage group were not included in the final model as they did not confound the association between race/ethnicity, SES, and disease-specific mortality.

Multivariable Cox proportional hazards regression also revealed increased adjusted all-cause mortality for similar sociodemographic and clinical groups (Table 4). Patients diagnosed at an older age (middle tertile 56-68 years: HR,

1.70; 95% CI, 1.44-2.01; upper tertile ≥69 years: HR, 3.32; 95% CI, 2.85-3.86; reference, lower tertile 0-55 years) and those treated with enucleation only (HR, 2.14; 95% CI, 1.88-2.44; reference, radiotherapy only) or both radiotherapy and enucleation (HR, 1.77; 95% CI, 1.12-2.80; reference, radiotherapy only) had increased HRs compared with their reference groups. The AJCC stage group confounded the association between race/ethnicity, SES, and all-cause mortality and was therefore included in this model, with more advanced stages being associated with increased mortality (stage 2: HR, 1.40; 95% CI, 1.19-1.65; stage 3: HR, 2.26; 95% CI, 1.87-2.73; and stage 4: HR, 10.09; 95% CI, 7.39-13.77; reference: stage 1). Sex did not confound the association between SES, race, ethnicity, and all-cause survival and was therefore excluded from the final model.

Discussion

This large, population-based study of patients with UM in the US highlights differences in treatment patterns and survival across racial/ethnic and socioeconomic groups. Given the limited difference in survival between radiotherapy and enucleation, radiotherapy is the preferred method to treat UM in appropriate circumstances, as outlined in the Collaborative Ocular Melanoma Study and recent National Comprehensive Cancer Network guidelines, allowing for globe salvage.⁴⁰⁻⁴⁵ Accordingly, enucleation is currently reserved as primary treatment for very large tumors, marked extrascleral extension, or UM

eyes with neovascular glaucoma or secondarily as salvage in eyes that do not respond to radiotherapy.^{26,46}

In this study, 20.1% of a nationally represented study population received primary enucleation; however, there was variation in treatment patterns, with primary enucleation more common in patients from nonwhite and lower socioeconomic backgrounds. While overall rates of primary enucleation decreased by 10.4 percentage points during the study period, nonwhite and socioeconomically disadvantaged patients consistently received primary enucleation in higher proportions than their white and socioeconomically advantaged counterparts. Although both disease-specific and all-cause 1-year and 5-year survival estimates had minimal racial/ethnic variation, lower SES was associated with decreased 5-year all-cause survival. However, the socioeconomic difference in survival was not large enough to show mortality differences, which suggests a difference in survival patterns but ultimately equivalent mortality outcomes. In addition, older age at diagnosis, advanced stage, and treatment with isolated enucleation were associated with increased mortality, supporting findings from previous reports.^{5,42} This study addressed knowledge gaps and suggests racial/ethnic and socioeconomic variation in treatment patterns and survival.

Understanding how and why patients with UM receive particular treatments and whether any racial/ethnic or socioeconomic disparities in care exist is important in this potentially fatal intraocular tumor.^{8-10,23} Previous research evaluating these factors in UM has focused mostly on incidence and diagnosis and less on treatment and survival.^{6,9,10,47-49} A predisposition for UM has been found among patients of white race.^{23,50-53} While these racial differences may be associated with the protective effects of darker skin pigmentation, it is acknowledged that racial/ethnic influences may also be associated with socioeconomic effects.^{7,54}

There are numerous hypotheses for a mechanism to explain the association between lower SES and increased likelihood of receiving primary enucleation. One hypothesis is that patients of lower SES face greater challenges receiving health care, thereby delaying their treatment and resulting in more advanced disease at presentation.²⁷ In this study, however, the AJCC stage at presentation was not associated with the patient's race, ethnicity, or SES status. An alternative hypothesis is that physicians change their management of patients with lower SES in an effort to enhance patient outcomes but find it difficult to balance what is feasible for a patient with the standard of care.⁵⁵ In patients with UM, such efforts could have the unintended consequence of increasing primary enucleation rates for patients of lower SES. Other potential confounding factors include patients' educational level, distance from centers offering globe-sparing treatments, and geographic variation in treatment patterns.^{56,57} Elucidating the mecha-

nisms through which SES may affect patient treatment and survival will require further research.

Limitations

This retrospective, population-based study has limitations. One limitation is that the preexisting SEER database used does not contain information about tumor size, visual acuity, follow-up visits, or individual patient-level SES. Accordingly, this study did not account for Collaborative Ocular Melanoma Study classification and tumor height. Previous research has shown the census tract-level composite SES to be a reasonable approximation for an individual's SES.³⁵ We were also unable to study the association of decreased access to health care because the SEER database does not include health services data. In addition, these data span a broad period of time dating back to 2004. Although several advances have been made in the field since then, this analysis of trends over time suggests that racial/ethnic and socioeconomic disparities have persisted despite changes in treatment patterns. In addition, because this was an observational study, these results represent associations as opposed to measures of causality.

Conclusions

The findings of this study suggest that, even in the post-Collaborative Ocular Melanoma Study era, primary enucleation continues to occur in disproportionate rates among patients from traditionally disadvantaged backgrounds. These racial/ethnic and socioeconomic differences have persisted despite overall treatment changes moving toward globe-sparing radiotherapy. Furthermore, these disparities appear to be unrelated to advanced disease at presentation, which may be suspected owing to delayed diagnosis in certain populations. In addition, while lower SES is associated with decreased all-cause survival, race/ethnicity did not appear to change survival rates.

Ophthalmologists other than ocular oncologists may also initially diagnose and increasingly manage UM. Counseling patients on treatment options may then occur at the point of initial contact with a variety of specialists. But in a time when globe-sparing therapies are successful and broadly available, our study suggests a need for further investigation into the reasons behind these disparities. Further investigation could elucidate the potential role that clinicians and variation in practice patterns play in these disparities. Given the consequences of enucleation, including postoperative vision loss, decreased quality of life, decreased functionality, and increased perceived stigma, efforts should be made to appropriately reduce differences in enucleation rates across populations.^{12,13,16}

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Invited Commentary

The Uneveled Field of Intraocular Malignancies

Jasmine H. Francis, MD

Socioeconomic, racial, and ethnic disparities are pervasive in medicine, and there is evidence for these differences within oncology practice. Disparities are associated with a complicated interplay between many factors, including biology, exposures, socioeconomic status (SES), race/ethnicity, culture, and health care systems.^{1,2} In many types of cancer, the story is the same: lower SES and nonwhite individuals have a higher incidence of cancer, which is diagnosed at a more advanced disease stage, and have higher cancer-specific mortality rates.^{1,2}

The reasons for racial, ethnic, and SES disparities in cancer mortality have been explained by factors relating to 3 categories: the tumor, patient characteristics, and health care.^{1,2} Regarding the tumor, nonwhite individuals, and patients of a

lower SES present with a later stage of cancer. It is unclear whether this differential is related to delays in diagnosis or to compromised access to care. In support of the latter, studies have suggested that in an equal-access health system, there was no difference in treatment delays.^{1,2} Differences in the quality and intensity of diagnostic investigation (eg, African American and Hispanic patients are less likely to undergo guideline-recommended positron emission or computed tomographic imaging at diagnosis of lung cancer³) may explain stage migration bias that occurs in deprived groups. The biology and molecular underpinnings of the tumor may also explain some of the differences, although this area has been relatively understudied for many cancers.

Regarding the patient, further investigations into how comorbidities and health-seeking behaviors influence treatment de-



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